



Ahmed Abdelghany

Senior Embedded Systems Engineer

Experienced software engineer with deep expertise in **embedded systems** using **C, C++, Python, and GIT**. With proven ability in solving complex problems.

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AREAS OF EXPERTISE

Real Time Operating Systems (RTOS)

Embedded Systems

Automotive Protocols

AutoSAR

C/C++

Data Structure & Algorithms

Software Development Lifecycle

COURSES

Algorithms and Systems for Autonomous Vehicles

Umeå university

2025 – on going

WORK EXPERIENCE

Senior Embedded Systems Engineer

Stockholm, SE

Scania (on Behalf of Afry)

Scania Special Vehicles Department

10/2025 – Present

- Investigated parallel and CAN display implementation and integration for a newly introduced ECU.
- Developed the HMI for a new in-vehicle monitor using **Qt (QML/Qt Quick)** and **C++14/17** on Embedded Linux, supporting **4-5 display applications**.
- Integrated CAN Display with the ECU by Modifying **C modules** for CAN protocol communication, data handling, and message parsing.

Scania Research and Development Asia Department

09/2024 – 09/2025

- Actively contributing to testing **ECUs** to ensure they align with safety, **cybersecurity**, and performance standards.
- Conducted vehicle and bench testing for **3+ different ECUs**, focusing on safety and cybersecurity requirements.
- Executed **10+ test cases** as outlined in the test specification and test plan to validate system functionality and performance.
- Coordinating and staying aligned with the team in China using **Jira** tickets for **tracking tasks, and issues**.

Senior C++ Software Engineer

Cairo, EG

Pharos Solution (Toyota Industries Corporation)

09/2023 - 08/2024

- Developed and maintained **C++14 code** for a comprehensive Warehouse Management System (WMS), focusing on enhancing automation and efficiency.
- Implemented new features in **C++**, analyzed legacy code, and **optimized system performance** through improved coding practices and algorithm optimization.
- Utilized **Qt** for developing user interfaces and **SQL databases** for managing data, ensuring robust and scalable solutions.
- Analyzed database structure and created effective SQL requests to streamline data operations.

Embedded System Engineer

Cairo, EG

The American University in Cairo (Ranked No. 1 University in Egypt)

06/2021 - 08/2023

- Monitored projects and research endeavors in the development of **C/C++ (14) applications**, applying V-Model, Scrum, and including **Open-Source AUTOSAR-compliant** Multicore Operating System, POSIX-Based Startup & Synchronization, and **Route Planner A* Algorithm**.
- Developed and Maintained Qt Application for Lab Management using **C++**, **SQL Database**.
- Teaching/Tutoring students on several topics/courses including **Embedded Systems, Object oriented programming**.

Embedded Systems Engineer

Cairo, EG

International Marathon United Technology (Military and Defense)

07/2018 - 05/2021

- Programmed Wheel Loader Control System to do various **motion control including Lift, Tilt**, and saving positions calculate bucket weight and **read input angle sensors** using **C/C++**
- Integrated ZF Transmission Control along with Cummins ECM and Vehicle control system **to read Input Speed Sensor**, Coolant **Temperature Sensor**, Coolant **Level Sensor**, and output **speed sensor**.
- Designed Software for Danfoss Joystick to control vehicle bucket using **CAN/J1939 Standards**.
- Configured **fan controller** to provide the most optimized airflow to transfer heat inside fluid to the air outside based on temperature sensors Including coolant and oil temperature sensors.

Software Engineer

Cairo, EG

Yodawy Healthcare

2016 - 2017

- Evaluated user requirements in collaboration with the product owner.
- Developed and designed creative mockups to assure user experience and flawless interaction.
- Developed XAML code to implement the designated Adobe Photoshop mockups

TECHNICAL SKILLS

SVN and GIT

POSIX

C++ 14 / 17

C/C++

Data Structure

Python

Bash

ASPICE Standard

Scrum/Agile

AUTOSAR

Linux

Adaptive AutoSAR

Object Oriented Programming

PROJECTS

Route Planner

Used **C++** to apply the **A* Algorithm combined** with OpenStreetMap data. This project utilized the IO2D visualization library, offering real-world route planning solutions.

POSIX-Based Startup & Synchronization

Developed a multi-process application using C Programming Language and POSIX fork() and execl(). The main process supervises child processes, relaying messages via anonymous pipes.

CERTIFICATES

C++ Developer Nanodegree

udacity.org

Certified Project Management Professional

pmi.org – Project Management Institute

IELTS Score of 7.0

ielts.idp.com - International English Language Testing System

Version Control with GIT

coursera.org

EDUCATION

9-Month Professional Diploma, Embedded Systems Track

Information Technology Institute

2017 - 2018

B.S. Electronics and Electrical Communications Engineering

Sohag University

2010 - 2015

LANGUAGES



Arabic



English



Swedish